

ST IGNATIUS COLLEGE RIVERVIEW



TASK 4

YEAR 12

2004

EXTENSION 2

TRIAL HSC EXAMINATION

Time allowed: 3 hours + 5 minutes reading time.

Instructions to Candidates

- Attempt **all** questions
- Show all necessary working.
- Marks may be deducted for missing or poorly arranged work.
- Board approved calculators may be used.
- Each question attempted must be returned in a *separate* writing booklet clearly marked Question 1, Question 2 etc, on the cover
- **Each booklet must have your name and the name of your mathematics teacher written on the cover.**

Question 1	{15 marks} Use a SEPARATE writing booklet.	Marks
a	If $Z_1 = 1 + 2i$, $Z_2 = 2 - i$ and $Z_3 = 1 - \sqrt{3}i$, Express in the form $(a + bi)$ where a and b are real. (i) $Z_1 + Z_2$ (ii) $\frac{1}{Z_2}$ (iii) $(Z_1)^3$	1 1 2
b	Express $\frac{4+3i}{3+i}$ in the form $(a + bi)$ where a and b are real numbers.	2
c	(i) Express $Z = \sqrt{3} + i$ in modulus- argument form. (ii) Hence, show that $Z^7 + 64Z = 0$.	1 3
d	(i) Find the square root(s) of $(-8 + 6i)$. (ii) Hence, solve the equation $2Z^2 - (3 + i)Z + 2 = 0$, expressing Z in the form $(a + bi)$ where a and b are real.	3 2

Question 2 (15 marks) Use a SEPARATE writing booklet.

Marks

a Evaluate

(i) $\int_0^{\frac{\pi}{4}} x \sin 2x \, dx$. 3

(ii) $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$. 2

(iii) $\int_0^{\frac{\pi}{3}} \frac{\tan x}{1+\cos x} \, dx$. (using $t = \tan \frac{x}{2}$). 4

b Show that , if $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$. 6

Then $I_n + I_{n-2} = \frac{1}{n-1}$, where n is an integer and $n \geq 3$

Hence evaluate I_7 .

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

a The point $A(a \cos \alpha, b \sin \alpha)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

B is the foot of the perpendicular from A to the x -axis. The normal at A cuts the x -axis at C .

(i) Represent this information with a suitable diagram.

1

(ii) Derive the equation of the normal AC .

3

(iii) Show that the length of CB is $\left| \frac{b^2 \cos \alpha}{a} \right|$.

3

b Consider the hyperbola H with equation $4x^2 - 9y^2 = 36$. The point $R(x_1, y_1)$ is an arbitrary point on H .

(i) Prove that the equation of the tangent l at R is $4x_1x - 9y_1y = 36$.

3

(ii) Find the co-ordinates of the point K at which l cuts the x -axis.

1

(iii) Hence, prove that $\frac{SR}{PR} = \frac{SK}{PK}$ where S and P are the foci of H .

4

Question 4 (15 marks) Use a SEPARATE writing booklet.

Marks

- a The equation $x^3 - 3x + 3 = 0$ has roots which are α, β and γ . Find the equation in x where the roots are α^2, β^2 and γ^2 . 4
- b The base of a solid is a circle of radius 2 units. A diameter runs through the centre of the base. Any cross section of the solid formed by a plane perpendicular to the given diameter is an equilateral triangle. 5
Show that the volume of the solid is $\frac{32\sqrt{3}}{3}$ units³.
- c The region bounded by the curve $y = \log_e x$, the straight lines $y = 1$ and $x = 3$ is rotated about the y -axis. Find the volume of the resulting solid using the method of cylindrical shells. 6

Question 5	(15 marks) Use a SEPARATE writing booklet.	Marks
a	Find the four fourth roots of -16 in the form $(a + bi)$.	4
b	A function is defined by $f(x) = \frac{\log_e x}{x}$ for $x > 0$.	
	(i) Find the x intercept.	1
	(ii) Find the turning point.	2
	(iii) Find the point of inflection.	2
	(iv) Sketch the graph of $y = f(x)$.	2
c	Consider the function in part (b) sketch	
i	$y = f(x) $.	2
ii	$y = \frac{1}{f(x)}$.	2

- a Consider the polynomial $Q(x) = ax^4 + bx^3 + cx^2 + dx + e$ where a, b, c, d and e are integers. Suppose α is an integer such that $Q(\alpha) = 0$.

(i) Prove that α is a factor of e .

2

(ii) Prove that the polynomial equation $P(x) = 0$,

2

where $P(x) = 4x^4 - x^3 + 3x^2 + 2x - 3$ does not have an integer root.

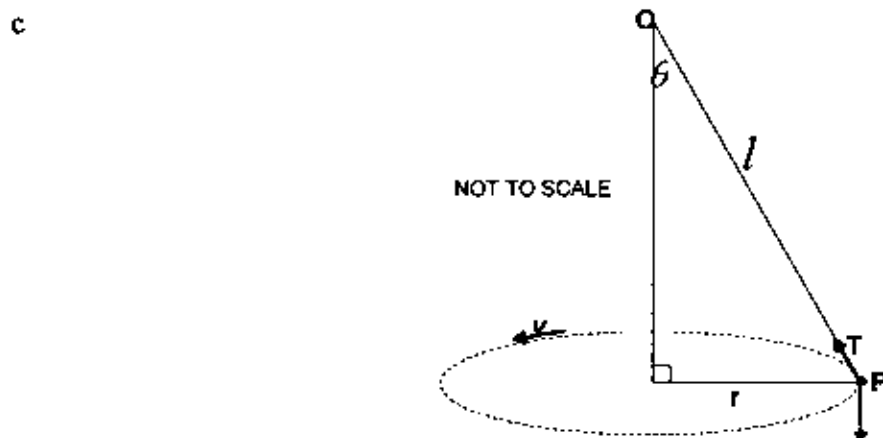
- b It is estimated that the probability that a torpedo will hit its target is $\frac{1}{3}$.

(i) If 5 torpedoes are fired, what is the probability of 3 successes.

2

(ii) How many torpedoes must be fired so that the probability of at least one success should be greater than 0.9?

2



The above diagram shows a light string of length l , fixed at O , and making an angle θ with the vertical as shown in the above diagram. A particle is attached at P . The particle moves with uniform speed v metres / second in a horizontal circle of radius r . The centre of the circle is directly below O .

If the particle is to maintain its motion in a horizontal circle, show by resolving forces vertically and horizontally, that the particle's velocity is given by

4

$v = \sqrt{rg \tan \theta}$. (Note: g is the acceleration due to gravity)

- d When a polynomial $P(x)$ is divided by $(x - 3)$ the remainder is 5 and when it is divided by $(x - 4)$ the remainder is 9. Find the remainder when $P(x)$ is divided by $(x - 4)(x - 3)$.

3

Question 7 (15 marks) Use a SEPARATE writing booklet. Marks

a If $\sin^{-1} x$, $\cos^{-1} x$ and $\sin^{-1}(1-x)$ are acute, show that 6

$$\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1.$$

Hence, solve the equation

$$\sin^{-1} x - \cos^{-1} x = \sin^{-1}(1-x).$$

b Find the general solution of the equation $3 \tan^2 x = 2 \sin x$. 5

c Each of the following statements is either true or false. Write 'True' or 'False' for each statement giving a brief reason for your answers. (You are not required to evaluate the integrals).

(i) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^3 \cos x \, dx = 0.$ 2

(ii) $\int_{-1}^1 e^{-x^2} \cos^{-1} x \, dx = 0.$ 2

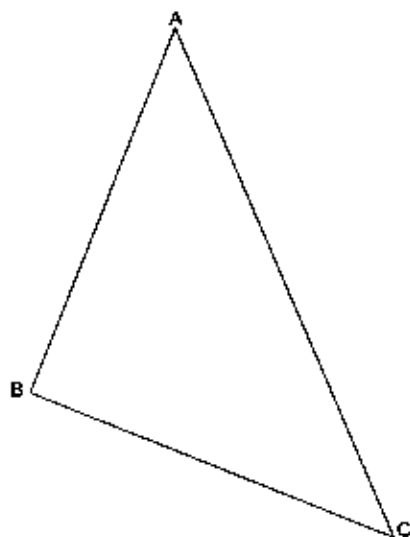
Question 8 (15 marks) Use a SEPARATE writing booklet.

Marks

- a In the Argand diagram, the points A , B and C represent the complex numbers Z_1 , Z_2 and Z_3 respectively.

What can you say about triangle ABC if $i(Z_3 - Z_2) = (Z_1 - Z_2)$.

2



- b Solve for x if $|3x + 3| + |x - 1| \leq 4x + 3$.

5

- c A particle, projected vertically upward with initial speed u is subjected to forces which create a constant vertical downward acceleration of magnitude g and an acceleration, directed against the motion, of magnitude kv when the speed is v .

8

(i) Show that the acceleration function is given by $\ddot{x} = -g - kv$.

(ii) Prove that the maximum height reached by the particle after a time T is given

$$\text{by } T = \frac{1}{k} \log_e \left(\frac{g + ku}{g} \right).$$

(iii) Prove that the maximum height reached is $\frac{1}{k}(u - gT)$.

(a) $z_1 = 1 + 2i$, $z_2 = 2 - i$ and $z_3 = 1 - \sqrt{3}i$

(1) (i) $z_1 + z_2 = 3 + i$

(1) (ii) $\frac{1}{z_2} = \frac{1}{2-i} \times \frac{(2+i)}{(2+i)}$
 $= \frac{2}{5} + \frac{1}{5}i$

(2) (iii) $(z_3)^3 = (1 - \sqrt{3}i)^3$
 $= 1 - 3\sqrt{3}i + 3(\sqrt{3}i)^2 - (\sqrt{3}i)^3$
 $= 1 - 3\sqrt{3}i - 9 + 3\sqrt{3}i$
 $= -8$

(2) (b) $\frac{4+3i}{3+i} \times \frac{(3-i)}{(3-i)} = \frac{12 - 4i + 9i + 3}{10}$
 $= \frac{3}{2} + \frac{1}{2}i$

(1) (c) (i) $z = \sqrt{3} + i$

$$\left. \begin{aligned} |z| &= \sqrt{3+1} \\ &= 2 \\ \arg z &= \tan^{-1} \frac{1}{\sqrt{3}} \\ &= \frac{\pi}{6} \end{aligned} \right\} z = 2 \operatorname{cis} \frac{\pi}{6}$$

$\therefore z = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$

(3) (ii) $z^7 = 2^7 \operatorname{cis} \frac{7\pi}{6}$ $64z = 64 \left(2 \operatorname{cis} \frac{\pi}{6} \right)$
 $= 128 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right)$ $= 128 \operatorname{cis} \frac{\pi}{6}$
 $= -128 \operatorname{cis} \frac{\pi}{6}$

$z^7 + 64z = 0$ (as required)

Question 1 (Continued)

(3) (d) (i) let $a + bi = \sqrt{-8 + 6i}$ ($a, b \in \mathbb{R}$)

$$a^2 - b^2 + 2abi = -8 + 6i$$

$$\therefore a^2 - b^2 = -8 \quad \text{--- (1)}$$

$$\text{and } 2ab = 6$$

$$ab = 3 \quad \text{--- (2)}$$

from (2) $a = \frac{3}{b}$ sub (1) : $\left(\frac{3}{b}\right)^2 - b^2 = -8$

$$9 - b^4 = -8b^2$$

$$b^4 - 8b^2 - 9 = 0$$

$$(b^2 - 9)(b^2 + 1) = 0$$

$$b^2 = -1 \quad \text{No Real sol}^n$$

$$b = \pm 3$$

when

$$b = 3 \quad a = 1$$

$$b = -3 \quad a = -1$$

The square roots are

$$1 + 3i \quad \text{and} \quad -1 - 3i$$

$$\text{or } \pm (1 + 3i)$$

(2) (ii)

$$2z^2 - (3+i)z + 2 = 0$$

$$z = \frac{(3+i) \pm \sqrt{(3+i)^2 - 4(2)(2)}}{4}$$

$$9 + 6i - 1 - 16$$

$$= \frac{(3+i) \pm \sqrt{-8+6i}}{4}$$

$$= \frac{3+i + 1+3i}{4}$$

$$\text{or } \frac{3+i - 1-3i}{4}$$

$$z = 1 + i$$

$$\text{or } z = \frac{1}{2} - \frac{1}{2}i$$

Question 2 (15 marks)

(3) (a) (i) $I = \int_0^{\frac{\pi}{4}} x \sin 2x \cdot dx$ let $u = x$ $\frac{dv}{dx} = \sin 2x$
 $\frac{du}{dx} = 1$ $V = -\frac{1}{2} \cos 2x$

$$= \left[-\frac{1}{2} x \cos 2x \right]_0^{\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} \frac{1}{2} \cos 2x \cdot dx$$

$$= \left[-\frac{\pi}{8} \cos \frac{\pi}{2} - 0 \right] + \frac{1}{4} \left[\sin 2x \right]_0^{\frac{\pi}{4}}$$

$$= 0 + \frac{1}{4} \left[\sin \frac{\pi}{2} - \sin 0 \right]$$

$$= \frac{1}{4}$$

(2) (ii) $I = \int_0^1 \frac{dx}{\sqrt{4-x^2}}$

$$= \left[\sin^{-1} \frac{x}{2} \right]_0^1$$

$$= \sin^{-1} \left(\frac{1}{2} \right) - \sin^{-1}(0)$$

$$= \frac{\pi}{6}$$

(4) (iii) $I = \int_0^{\frac{\pi}{3}} \frac{\tan x}{1 + \cos x} \cdot dx$

$$I = \int_0^{\frac{1}{\sqrt{3}}} \frac{\frac{2t}{1-t^2}}{1 + \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2t}{1-t^2} \times \frac{(1+t^2)}{2} \times \frac{2}{(1+t^2)} \cdot dt$$

let $t = \tan \frac{x}{2}$
 $x = 2 \tan^{-1} t$
 $\frac{dx}{dt} = \frac{2}{1+t^2}$

when $x = 0$ $t = 0$
 when $x = \frac{\pi}{3}$ $t = \frac{1}{\sqrt{3}}$

Question 2 (Continued)

(a) (iii)
$$I = \int_0^{\frac{1}{\sqrt{3}}} \frac{2t}{1-t^2} \cdot dt$$

$$= - \left[\ln(1-t^2) \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= - \left[\ln \frac{2}{3} - \ln 1 \right]$$

$$= \ln \frac{3}{2}$$

(b)
$$I_n = \int_0^{\frac{\pi}{4}} \tan^n x \cdot dx$$

$$I_n + I_{n-2} = \int_0^{\frac{\pi}{4}} \tan^n x \cdot dx + \int_0^{\frac{\pi}{4}} \tan^{n-2} x \cdot dx$$

(3)
$$= \int_0^{\frac{\pi}{4}} \tan^{n-2} x \cdot (\tan^2 x + 1) \cdot dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^{n-2} x \cdot \sec^2 x \cdot dx$$

$$= \frac{1}{n-1} \left[\tan^{n-1} x \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{n-1} \left[\left(\tan \frac{\pi}{4} \right)^{n-1} - 0 \right]$$

$$= \frac{1}{n-1} \quad (\text{as required})$$

let $u = \tan x$

$$\frac{du}{dx} = \sec^2 x$$

$$du = \sec^2 x \cdot dx$$

$$\int u^{n-2} \cdot du$$

$$= \frac{1}{n-1} \cdot u^{n-1}$$

Question 2 (Continued)

(3) (b) Using $I_n + I_{n-2} = \frac{1}{n-1}$, $n \geq 3$

$$n=7 \quad \text{Then} \quad I_7 + I_5 = \frac{1}{6}$$

$$I_7 = \frac{1}{6} - I_5$$

$$n=5 \quad \text{Then} \quad I_5 + I_3 = \frac{1}{4}$$

$$I_5 = \frac{1}{4} - I_3$$

$$n=3 \quad \text{Then} \quad I_3 + I_1 = \frac{1}{2}$$

$$I_1 = \int_0^{\frac{\pi}{4}} \tan x \, dx$$

$$= - \left[\ln \cos x \right]_0^{\frac{\pi}{4}}$$

$$= - \left[\ln \frac{1}{\sqrt{2}} - \ln 1 \right]$$

$$= \ln \sqrt{2} \quad \text{or} \quad \frac{1}{2} \ln 2$$

$$\begin{aligned} \tan x &= \frac{\sin x}{\cos x} \\ &= \frac{f'(x)}{f(x)} \end{aligned}$$

$$I_7 = \frac{1}{6} - \frac{1}{4} + \frac{1}{2} - \frac{1}{2} \ln 2$$

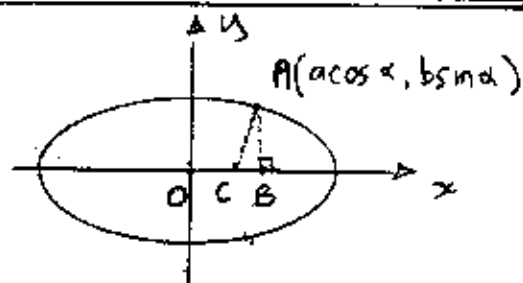
$$\frac{2}{12} - \frac{3}{12} + \frac{6}{12}$$

$$= \frac{5}{12} - \frac{1}{2} \ln 2$$

$$\approx 0.07$$

Question 3

(1) (a) (i)



(3)

(ii)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{by implicit diff}^n \text{ we have}$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$$

$$m_T = -\frac{b^2 x}{a^2 y}$$

$$m_N = \frac{a^2 y}{b^2 x}$$

Equation of Normal at

A (a cos alpha, b sin alpha)

$$y - b \sin \alpha = \frac{a^2 b \sin \alpha}{b a \cos \alpha} (x - a \cos \alpha)$$

$$(b \cos \alpha) y - b^2 \sin \alpha \cos \alpha = a \sin \alpha (x - a \cos \alpha)$$

$$= (a \sin \alpha) x - a^2 \sin \alpha \cos \alpha$$

$\div$ by $\sin \alpha \cos \alpha$

$$\frac{by}{\sin \alpha} - b^2 = \frac{ax}{\cos \alpha} - a^2$$

$$\frac{2}{a} - b^2 = \frac{ax}{\cos \alpha} - \frac{by}{\sin \alpha}$$

(3)

(iii)

put $y=0$

$$\frac{2}{a} - b^2 = \frac{ax}{\cos \alpha}$$

$$x = \frac{\cos \alpha (a^2 - b^2)}{a}$$

Question 3 (Continued)

(a)

$$CB = |OB - OC|$$

(iii)

$$\begin{aligned} &= \left| a \cos \alpha - \frac{a^2 - b^2}{a} \cos \alpha \right| \\ &= \cos \alpha \left[\frac{a^2 - (a^2 - b^2)}{a} \right] \\ &= \left| \frac{b^2 \cos \alpha}{a} \right| \quad (\text{as required}) \end{aligned}$$

(b)

$$4x^2 - 9y^2 = 36$$

$R(x_1, y_1)$ on Hyperbola

(3) (i)

By implicit differentiation

$$8x - 18y \cdot \frac{dy}{dx} = 0$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{8x}{18y} \\ &= \frac{4x}{9y} \end{aligned}$$

Equation of
tangent at
 $R(x_1, y_1)$

$$\begin{aligned} y - y_1 &= \frac{4x_1}{9y_1} (x - x_1) \\ 9y_1 y - 9y_1^2 &= 4x_1 x - 4x_1^2 \end{aligned}$$

Since $R(x_1, y_1)$ lies on H then $4x_1^2 - 9y_1^2 = 36$

$$36 = 4x_1 x - 9y_1 y \quad (\text{as required})$$

(ii)

Question 3 (Continued)

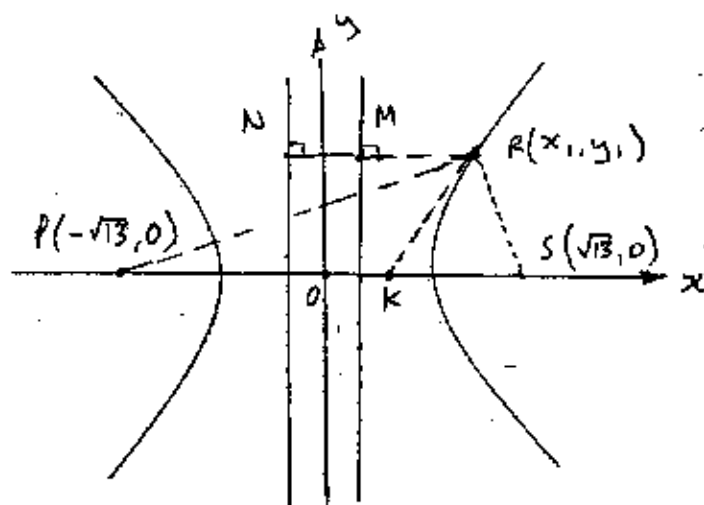
(1) (b) (ii) It cuts x-axis when $y = 0$

$$36 = 4x, x = 9$$

$$x = \frac{9}{x_1}$$

$$\therefore K\left(\frac{9}{x_1}, 0\right)$$

(4) (ii) Prove $\frac{SR}{PR} = \frac{SK}{PK}$



$$SK = OS - OK \quad PK = OP + OK$$

$$= \sqrt{13} - \frac{9}{x_1} \quad = \sqrt{13} + \frac{9}{x_1}$$

$$\frac{SK}{PK} = \frac{\sqrt{13} - \frac{9}{x_1}}{\sqrt{13} + \frac{9}{x_1}}$$

$$= \frac{x_1 \sqrt{13} - 9}{x_1 \sqrt{13} + 9}$$

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$a = 3$$

$$b = 2$$

$$b^2 = a^2(e^2 - 1)$$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$= 1 + \frac{4}{9}$$

$$e = \frac{\sqrt{13}}{3}$$

$$S(\pm ae, 0)$$

$$S(\pm \sqrt{13}, 0)$$

$$x = \pm \frac{a}{e}$$

$$x = \pm \frac{9}{\sqrt{13}}$$

Question 3 (Continued)

(b) (ii) By defⁿ of Hyperbola, we have

$$\begin{array}{l} \text{and} \quad \left. \begin{array}{l} \frac{SR}{MR} = e \Rightarrow SR = eMR \\ \frac{PR}{NR} = e \Rightarrow PR = eNR \end{array} \right\} \frac{SR}{PR} = \frac{MR}{NR} \end{array}$$

$$MR = x_1 - \frac{9}{\sqrt{13}} \quad \text{and} \quad NR = x_1 + \frac{9}{\sqrt{13}}$$

$$\frac{MR}{NR} = \frac{x_1 - \frac{9}{\sqrt{13}}}{x_1 + \frac{9}{\sqrt{13}}}$$

$$\therefore \frac{SR}{PR} = \frac{x_1 \sqrt{13} - 9}{x_1 \sqrt{13} + 9}$$

$$= \frac{SK}{PK} \quad (\text{as required})$$

Question 4 (15 marks)

(4) (a)

$$x^3 - 3x + 3 = 0$$

roots α, β and γ

α^2, β^2 and γ^2 will satisfy

$$x^{\frac{3}{2}} - 3x^{\frac{1}{2}} + 3 = 0$$

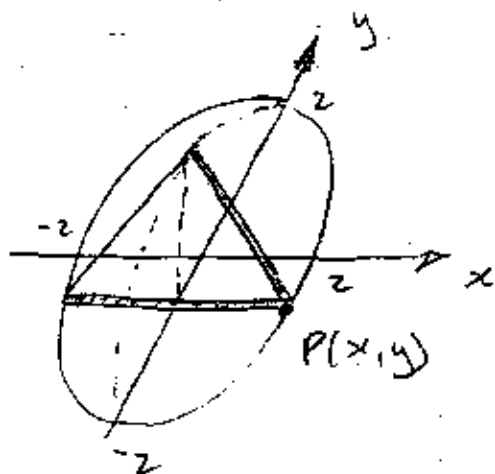
$$\text{i.e. } x^{\frac{1}{2}}(x - 3) = -3$$

$$\text{Squaring: } x(x - 3)^2 = 9$$

$$x(x^2 - 6x + 9) = 9$$

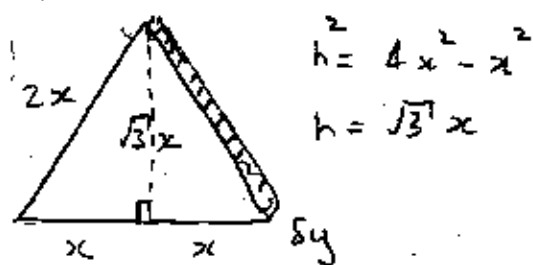
$$x^3 - 6x^2 + 9x - 9 = 0$$

(5) (b)



$$\text{Base } x^2 + y^2 = 4$$

Typical cross section



Typical Volume

$$\delta V = \sqrt{3}x^2 \cdot \delta y$$

$$V = \int_{-2}^2 \sqrt{3}x^2 \cdot dy$$

$$= 2\sqrt{3} \int_0^2 (4 - y^2) \cdot dy$$

$$= 2\sqrt{3} \left[4y - \frac{y^3}{3} \right]_0^2$$

$$= 2\sqrt{3} \left[8 - \frac{8}{3} \right]$$

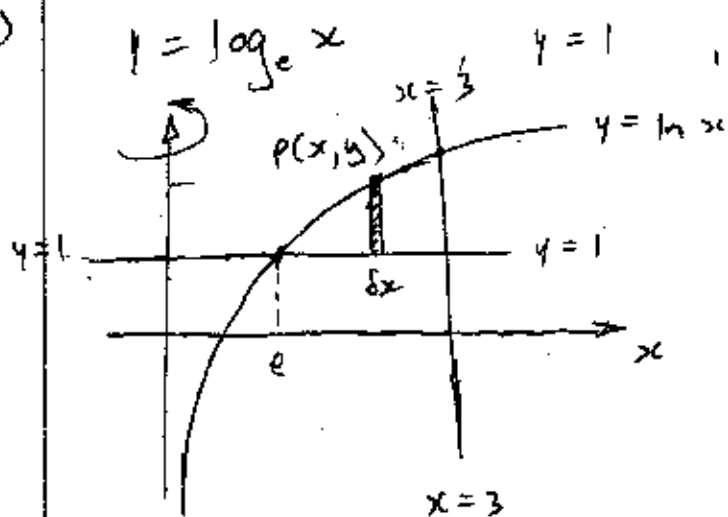
$$\begin{aligned} \delta A &= \frac{1}{2}bh \\ &= \frac{1}{2} \cdot 2x \cdot \sqrt{3}x \\ &= \sqrt{3}x^2 \end{aligned}$$

$$\text{Using } x^2 = 4 - y^2$$

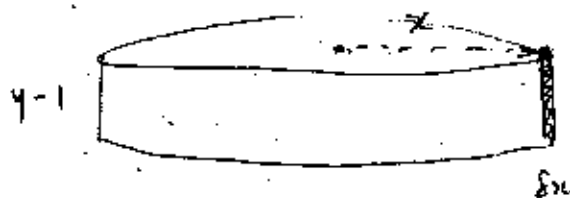
$$= \frac{32\sqrt{3}}{3} \text{ cubic units}$$

Question 4 (Continued)

(6) (c)



Typical Shell



$$\delta V = 2\pi x (y-1) \cdot \delta x$$

$$= 2\pi x (\ln x - 1) \cdot \delta x$$

$$V = 2\pi \int_e^3 (x \ln x - x) \cdot dx$$

L.I.A.T.E.

$$\frac{V}{2\pi} = \int_e^3 x \ln x \cdot dx - \left[\frac{x^2}{2} \right]_e^3$$

Integ. By Parts

$$u = \ln x \quad \frac{dv}{dx} = x$$

$$\frac{du}{dx} = \frac{1}{x} \quad v = \frac{1}{2} x^2$$

$$= \left[\frac{1}{2} x^2 \ln x \right]_e^3 - \int_e^3 \frac{1}{2} x^2 \cdot \frac{1}{x} \cdot dx - \left[\frac{x^2}{2} \right]_e^3$$

$$= \frac{9}{2} \ln 3 - \frac{1}{2} e^2 - \left[\frac{x^2}{4} \right]_e^3 - \left[\frac{x^2}{2} \right]_e^3$$

$$= \frac{9}{2} \ln 3 - \frac{1}{2} e^2 - \left[\frac{3x^2}{4} \right]_e^3$$

$$= \frac{9}{2} \ln 3 - \frac{1}{2} e^2 - \frac{27}{4} + \frac{3e^2}{4}$$

$$V = \pi \left[9 \ln 3 - e^2 - \frac{27}{2} + \frac{3e^2}{2} \right]$$

$$= \pi \left[9 \ln 3 - \frac{27}{2} + \frac{e^2}{2} \right] \text{ units}^3$$

Question 5 (15 marks)

(4) (a) let $z = r \operatorname{cis} \theta$

where $z^4 = -16$

$$r^4 \operatorname{cis} 4\theta = 16 \operatorname{cis} \pi$$

$$\therefore r = 2 \quad \text{and} \quad 4\theta = \pi + 2n\pi, \quad n \in \mathbb{Z}$$

$$\theta = \frac{\pi + 2n\pi}{4}$$

$$z_1 = 2 \operatorname{cis} \frac{\pi}{4} = 2 \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right) = \sqrt{2} + \sqrt{2}i$$

$$z_2 = 2 \operatorname{cis} \frac{3\pi}{4} = -\sqrt{2} + \sqrt{2}i$$

$$z_3 = 2 \operatorname{cis} \frac{5\pi}{4} = -\sqrt{2} - \sqrt{2}i$$

$$z_4 = 2 \operatorname{cis} \frac{7\pi}{4} = \sqrt{2} - \sqrt{2}i$$

(b) $f(x) = \frac{\ln x}{x}$, for $x > 0$

(1) (i) let $f(x) = 0$: $\frac{\ln x}{x} = 0$
 $x = 1$

(2) (i) $f'(x) = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$
 $= \frac{1 - \ln x}{x^2}$

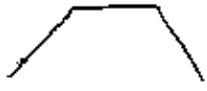
x	2	e	3
$f'(x)$	+	0	-

put $f'(x) = 0$: $\frac{1 - \ln x}{x^2} = 0$

$$\ln x = 1$$

$$x = e$$

$$\text{and } f(e) = \frac{1}{e}$$


 Max T.P. at
 $\left(e, \frac{1}{e}\right)$

Question 5 (continued)

(b) (iii) $f'(x) = \frac{1 - \ln x}{x^2}$

(2) $f''(x) = \frac{x^2 \cdot -\frac{1}{x} - (1 - \ln x) \cdot 2x}{x^4}$
 $= \frac{-x - 2x + 2x \ln x}{x^4}$
 $= \frac{2 \ln x - 3}{x^3}$

When $f''(x) = 0$: $2 \ln x - 3 = 0$

$$\ln x = \frac{3}{2}$$

$$x = e^{\frac{3}{2}}$$

When $x = e^{\frac{3}{2}}$: $y = \frac{\frac{3}{2}}{e^{\frac{3}{2}}}$

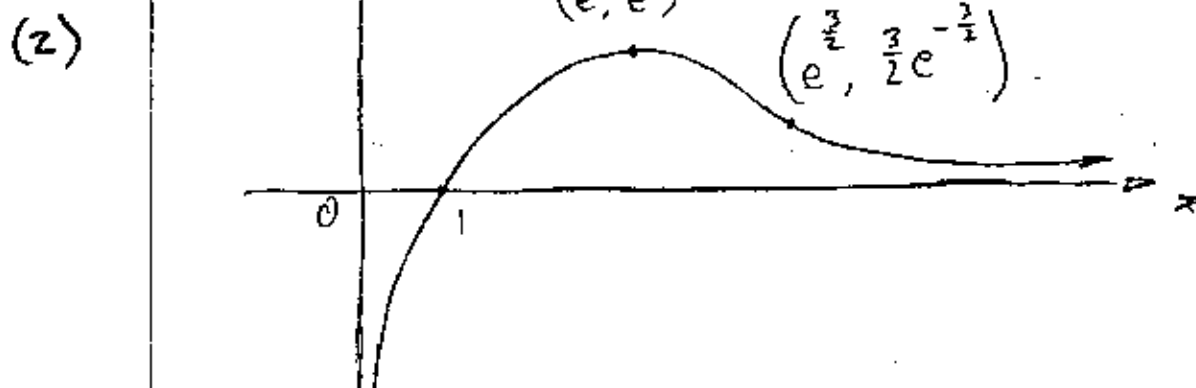
Concavity Test

x	e	$e^{\frac{3}{2}}$	e^2
$f''(x)$	-	0	+

$$= \frac{3}{2e^{\frac{3}{2}}}$$

$\therefore$ I.P. at $\left(e^{\frac{3}{2}}, \frac{3}{2e^{\frac{3}{2}}}\right)$

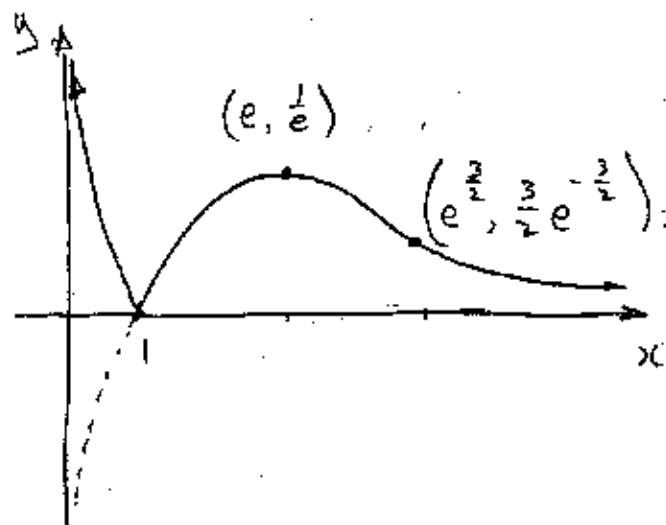
(iv) as $x \rightarrow \infty$ $\frac{\ln x}{x^2} \rightarrow 0$ (Since x dominates $\ln x$)



Question 5 (Continued)

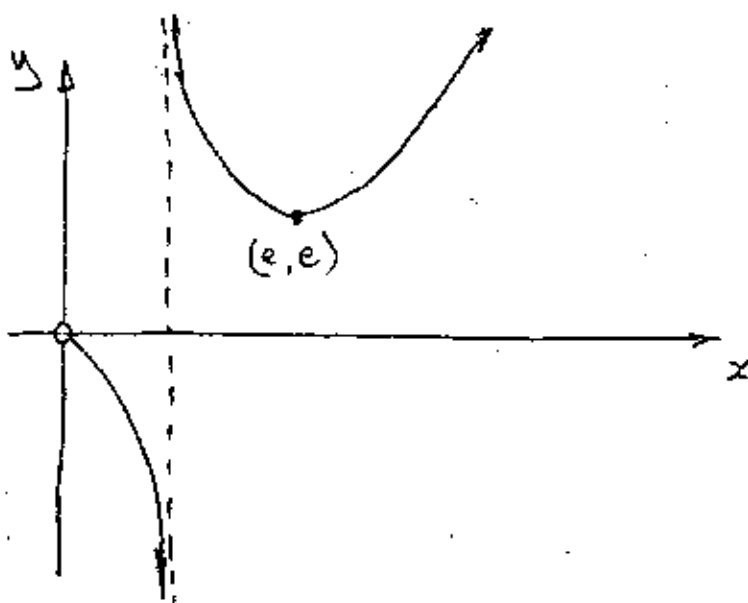
(a) (i) $y = |f(x)|$

(2)



(ii) $y = \frac{1}{f(x)}$

(2)



Question 6 (15 marks)

(a) $Q(x) = ax^4 + bx^3 + cx^2 + dx + e$

(2) (i) given α is an integer : $Q(\alpha) = 0$

$$Q(\alpha) = a\alpha^4 + b\alpha^3 + c\alpha^2 + d\alpha + e$$

letting $Q(\alpha) = 0$: $a\alpha^4 + b\alpha^3 + c\alpha^2 + d\alpha = -e$
 $\alpha(a\alpha^3 + b\alpha^2 + c\alpha + d) = -e$

Since a, b, c, d are integers then $k\alpha = -e$
 where k is also an integer

hence α is a factor of e

(2) (i) $P(x) = 4x^4 - x^3 + 3x^2 + 2x - 3$

The only possible integer roots are ± 1 and ± 3

$$P(1) = 4 - 1 + 3 + 2 - 3 \neq 0$$

$$P(-1) = 4 + 1 + 3 - 2 - 3 \neq 0$$

$$P(3) = 324 - 27 + 27 + 6 - 3 \neq 0$$

$$P(-3) = 324 + 27 + 27 - 6 - 3 \neq 0$$

$\therefore P(x) = 0$ does not have an integer root

Five Torpedoes fired

(2) (i) $P(H) = \frac{1}{3}$
 $P(M) = \frac{2}{3}$ } $P(3 \text{ Hits}) = {}^5C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^2$
 $= \frac{40}{243}$

(2) (ii) $P(\text{at least 1 hit}) = 1 - P(\text{no hits})$
 $= 1 - \left(\frac{2}{3}\right)^5$
 $\therefore 1 - \left(\frac{2}{3}\right)^5 > 0.9$

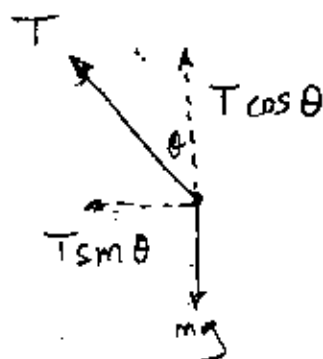
$\left(\frac{2}{3}\right)^n < 0.1$
 $\left(\frac{2}{3}\right)^5 = 0.132$
 $\left(\frac{2}{3}\right)^6 = 0.088$

Requires 6 torpedoes

Question 6 (Continued)

(c)

Forces on P



Vertically (Zero net force)

$$T \cos \theta = mg \quad - (1)$$

Radially

$$T \sin \theta = \frac{mv^2}{r} \quad - (2)$$

$$(2) \div (1) : \quad \tan \theta = \frac{\frac{mv^2}{r}}{mg}$$

$$= \frac{v^2}{gr}$$

$$v^2 = gr \tan \theta$$

$$v = \sqrt{gr \tan \theta} \quad (\text{as required})$$

(d)

$$P(x) = (x-3)(x-4) + R(x)$$

(3) Degree of $R(x) < 2$

$$\text{let } R(x) = ax + b$$

$$P(3) = 5 : \quad 5 = 3a + b \quad - (1)$$

$$P(4) = 9 : \quad 9 = 4a + b \quad - (2)$$

$$(2) - (1) : \quad 4 = a$$

$$\text{sub } (2) \quad 9 = 16 + b$$

$$b = -7$$

$$\therefore R(x) = 4x - 7$$

Question 7 (15 marks)

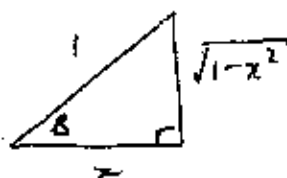
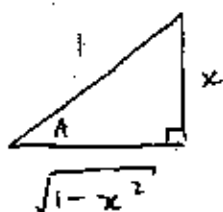
(a) $\sin^{-1} x$, $\cos^{-1} x$ and $\sin^{-1}(1-x)$ are acute

let $A = \sin^{-1} x$ let $B = \cos^{-1} x$

(3)

$$\sin A = x$$

$$\cos B = x$$



$$\begin{aligned} \sin(A-B) &= \sin A \cos B - \cos A \sin B \\ &= x^2 - \sqrt{1-x^2} \cdot \sqrt{1-x^2} \\ &= x^2 - (1-x^2) \\ &= 2x^2 - 1 \quad (\text{as required}) \end{aligned}$$

(3)

Solve $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(1-x)$

$$\sin(A-B) = \sin(\sin^{-1}(1-x))$$

$$2x^2 - 1 = 1 - x$$

$$2x^2 + x - 2 = 0$$

$$x = \frac{-1 \pm \sqrt{1 + 4(2)(2)}}{4}$$

$$= \frac{-1 \pm \sqrt{17}}{4}$$

Since $x = \sin A$
then $-1 \leq x \leq 1$

$$x = \frac{-1 + \sqrt{17}}{4}$$

$$(\approx 0.78)$$

Question 7. (continued)

(b) $3 \tan^2 x = 2 \sin x$

(5) $3 \frac{\sin^2 x}{\cos^2 x} = 2 \sin x$

$3 \sin^2 x = 2 \sin x (1 - \sin^2 x)$

$3 \sin^2 x = 2 \sin x - 2 \sin^3 x$

$2 \sin^3 x + 3 \sin^2 x - 2 \sin x = 0$

$\sin x (2 \sin^2 x + 3 \sin x - 2) = 0$

$\sin x (2 \sin x - 1)(\sin x + 2) = 0$

$$\begin{array}{r} 2 \sin x & -1 \\ \sin x & +2 \end{array}$$

either $\sin x = 0$ or $\sin x = \frac{1}{2}$ $\sin x \neq -2$

$x = n\pi$ $x = \frac{\pi}{6}, \frac{5\pi}{6}, \dots$

$x = n\pi + (-1)^n \cdot \frac{\pi}{6}$

(c) (i) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^3 \cos x \cdot dx = 0$ True

(2) $\left. \begin{array}{l} -x^3 \text{ - is an odd function} \\ \cos x \text{ - is an even function} \end{array} \right\} \text{odd} \times \text{even} = \text{odd}$

(ii) $\int_{-1}^1 e^{-x^2} \cos^{-1} x \cdot dx = 0$ False

(2) $\left. \begin{array}{l} \cos^{-1} x > 0 \text{ for } -1 < x < 1 \\ e^{-x^2} > 0 \text{ for all } x \end{array} \right\}$

$\therefore$ product can never equal zero

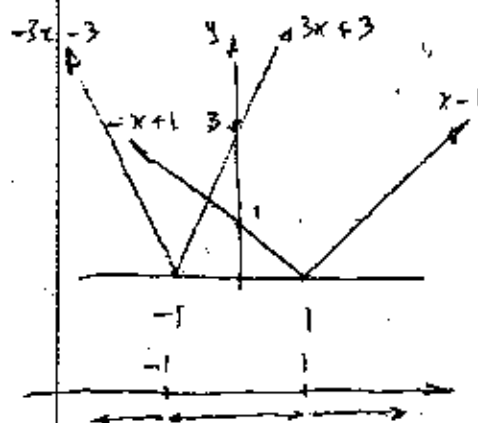
Question 8 (15 marks)

4x+3

(b)

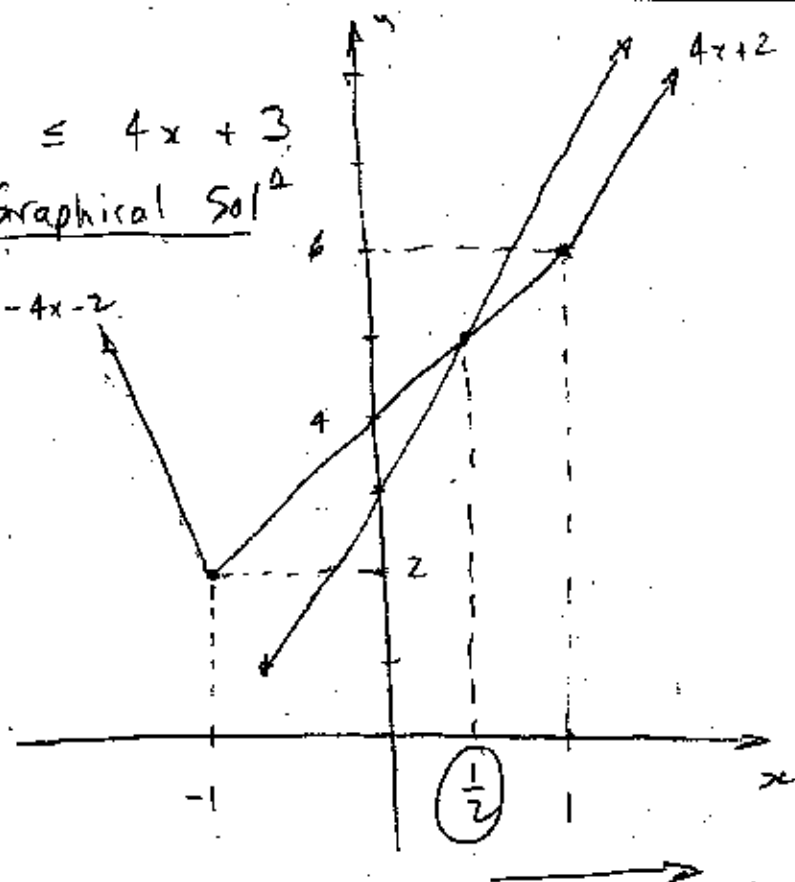
$$|3x+3| + |x-1| \leq 4x+3$$

Graphical Solⁿ



(5)

$$-4x-2$$



Graphical Solⁿ

$$x \geq \frac{1}{2}$$

Algebraic Solⁿ

for $x \leq -1$

$$-4x-2 \leq 4x+3$$

$$-8x \leq 5$$

$$x \geq -\frac{5}{8}$$

No Solⁿ

for $-1 \leq x \leq 1$

$$2x+4 \leq 4x+3$$

$$1 \leq 2x$$

$$x \geq \frac{1}{2}$$

$$\therefore \frac{1}{2} \leq x \leq 1$$

for $x \geq 1$

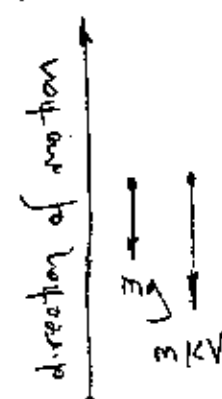
$$4x+2 \leq 4x+3$$

$$2 \leq 3$$

True for all $x \geq 1$

$$\text{Final Solⁿ: } x \geq \frac{1}{2}$$

ve x



$$t=0$$

$$y=0$$

$$x=0$$

(i) Forces on P

$$m\ddot{x} = -mg - mkv$$

$$\ddot{x} = -g - kv$$

(ii)

$$\ddot{x} = -(g + kv)$$

$$\frac{dv}{dt} = -(g + kv)$$

$$\frac{dt}{dv} = \frac{-1}{g + kv}$$

Question 8 (continued)

$$(b) \quad t = \int \frac{-1}{g + kv} \cdot dv$$

$$= -\frac{1}{k} \ln(g + kv) + C$$

when $t = 0$
 $V = u$

 $\therefore C = \frac{1}{k} \ln(g + ku)$

$$t = \frac{1}{k} \ln \left(\frac{g + ku}{g + kv} \right)$$

max height
 when $V = 0$

$$T = \frac{1}{k} \ln \left(\frac{g + ku}{g} \right) \quad (\text{as required})$$

$$(ii) \quad \text{Using } \ddot{x} = -(g + kv)$$

$$(4) \quad v \cdot \frac{dv}{dx} = -(g + kv)$$

$$\frac{dv}{dx} = -\left(\frac{g + kv}{v}\right)$$

$$\frac{dx}{dv} = -\frac{v}{g + kv}$$

$$= -\frac{1}{k} \left(\frac{kv + g - g}{g + kv} \right)$$

$$= -\frac{1}{k} \left(1 - \frac{g}{g + kv} \right)$$

$$x = -\frac{1}{k} \int \left(1 - \frac{g}{g + kv} \right) \cdot dv$$

$$= -\frac{1}{k} \left[v - \frac{g}{k} \ln(g + kv) \right] + C$$

when $x = 0$
 $V = u$

$$C = \frac{1}{k} \left[u - \frac{g}{k} \ln(g + ku) \right]$$

Question 8 (Continued)

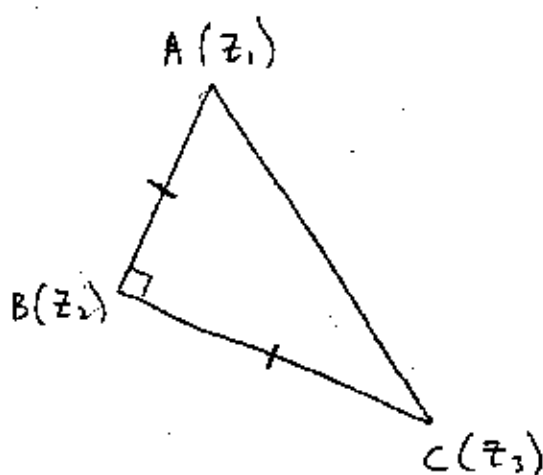
(b) $x = \frac{1}{k} \left[(u-v) - \frac{g}{k} \ln \left(\frac{g+ku}{g+kv} \right) \right]$

Max. height
when $v=0$

$$x = \frac{1}{k} \left[u - \frac{g}{k} \ln \left(\frac{g+ku}{g} \right) \right]$$

$$= \frac{1}{k} [u - gT] \quad (\text{as required})$$

(c)
(2)



Note $i(z_3 - z_2)$ rotates
the vector by 90°
(anti-clockwise)

ABC is a right angled triangle with $AB = BC$

$$|z_3 - z_2| = BC \quad \text{and} \quad |z_1 - z_2| = AB$$